

# CODING & ROBOTICS – GRADE 4 STUDY NOTES & CODE

## TERM 1 CONTENT

### LESSON 7 Control in Scratch

- Repeat loops in our daily lives.
- Use a wait block in Scratch.

#### CODE – REPEAT LOOP

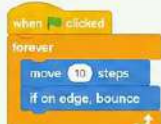


**KEY:** repeat – do something again and again.

### LESSON 9 Loops and conditional statements

- Forever loops in real-life situations.
- If – then – else statements in real-life situations.

#### CODE – FOREVER LOOP



**KEY:** forever – keep doing something without stopping.

#### CODE – IF THEN ELSE



**KEY:** if – then – else – make decisions.

### LESSON 10 Our digital world

- What is technology? (Technology, non-technology and IT).
- Input, Processing and Output (IPO).

#### IPO CYCLE



#### EXAMPLES

- Keyboard (input) → Computer (processing) → Screen (output)
- Camera (input) → Phone (processing) → Photo (output)

## TERM 2 CONTENT

### LESSON 11 Sensing in Scratch

- Colour-sensing blocks.

#### CODE – COLOUR SENSING



**KEY:** Detect colours and respond to them.

### LESSON 12 Variables and Operators

- What are variables?
- What are operators?

#### CODE – VARIABLE



#### CODE – OPERATORS



### LESSON 14 AND 15 Use the pen function.

- What are pens, why do we need them, etc.
- Pen code blocks.
- Use the pen function in Scratch.
- Draw a hexagon in Scratch.

#### PEN CODE BLOCKS



#### CODE – DRAW HEXAGON



### LESSON 16 Robots and their impact on the world.

- What is a robot, what job does it do, etc.

#### EXAMPLES OF ROBOTS

- Industrial Robot – used in factories.
- Medical Robot – used in surgeries.
- Domestic Robot – used for cleaning, etc.
- Space Robot – used for exploration.

### REAL-LIFE CONNECTIONS

- Loops – traffic lights, washing machine cycle.
- If – Then – Else – crossing the road, weather decisions.
- IPO – recipe (ingredients → cooking → food).
- Variables – keeping score in a game.
- Pen Function – drawing shapes and patterns.
- Robots – help humans in many jobs.

### EXAM TIPS

- Read the question carefully.
- Use correct Scratch blocks.
- Test your code before submitting.
- Use meaningful variable names.
- Check your project and save your work.

### PRACTICE IDEAS

- Make a sprite move in a square using a loop.
- Use if – then – else to change costume.
- Make a colour-sensing game.
- Keep score using a variable.
- Draw different shapes using the pen.
- Research a robot and write its uses.

| KEY VOCABULARY           |                                                  |
|--------------------------|--------------------------------------------------|
| <b>Loop:</b>             | A block of code that repeats.                    |
| <b>Repeat Loop:</b>      | Repeats a block of code a fixed number of times. |
| <b>Forever Loop:</b>     | Repeats forever (until stopped).                 |
| <b>Wait Block:</b>       | Pauses the code for some time.                   |
| <b>If – Then – Else:</b> | A conditional statement used to make decisions.  |
| <b>Technology:</b>       | Things that solve problems and make life easier. |
| <b>IT:</b>               | Information Technology.                          |
| <b>Input:</b>            | Data entered into a device.                      |
| <b>Processing:</b>       | Data is worked on.                               |
| <b>Output:</b>           | Information produced.                            |
| <b>Variable:</b>         | A storage box for data.                          |
| <b>Operator:</b>         | A symbol used to perform operations.             |
| <b>Pen Function:</b>     | Used to draw with the sprite.                    |
| <b>Robot:</b>            | A machine that can do tasks automatically.       |

## SCRATCH CODE QUICK REFERENCE

|                  |                            |
|------------------|----------------------------|
| <b>Events</b>    | when green flag clicked    |
| <b>Control</b>   | repeat 10 times, forever   |
| <b>Sensing</b>   | touching edge?             |
| <b>Operators</b> | 5 + 3, pick random 1 to 10 |
| <b>Variables</b> | set score to 0             |
| <b>Pen</b>       | pen down                   |

### ★ IMPORTANT REMEMBER

- Practice regularly.
- Understand each concept.
- Use code blocks correctly.
- Think logically and be creative!

KEEP PRACTICING, KEEP CODING!

